★ SECTION 1: CORE INFRASTRUCTURE
★ SECTION 3: SCHEDULING & FARE
★ SECTION 5: OPERATIONS — Delay, Incident, Tracking, Alert, Maintenance
★ SECTION 6: BUS TICKETING (uses metro.PASSENGER)
★ SECTION 7: METRO ↔ BUS INTEGRATION (KEY SECTION)
★ SECTION 8: BUS FEEDBACK + SYSTEM CONFIG
LEGEND — Chen's Notation
Strong Entity
Weak Entity
Relationship
Identifying Rel.
Attribute
Key Attribute
Derived Attr (dashed)
metro.STATION (external)
metro.PASSENGER (external)
Red = External metro schema reference
== double line = Total participation
— single line = Partial participation
BUS_OPERATOR
operator_name
operator_id
operator_type
city
operator_code
is_active
BUS_ROUTE
route_number
route_id
route_type
route_name
is_active
total_length_km
OPERATES
OWNED_BY
ROUTE_STOP
sequence_no
is_active
is_terminal
dist_from_start_km
BUS_STOP
stop_code
stop_id
longitude
latitude
stop_name
stop_type
city
has_shelter
is_active
zone_no
BUS
bus_type
bus_number
manufacturer
bus_id
current_crowd_count
current_lat
current_lng
total_capacity
operational_status
ASSIGNED_TO
BUS_DRIVER
license_no
full_name
driver_id
salary
experience_years
employment_status
phone
DRIVES
BUS_DRIVER_ASSIGNMENT
assignment_date
role
FOR_SCHEDULE
ASSIGNED_SHIFT
BUS_DRIVER_SHIFT
working_hours
shift_date
shift_id
shift_end
shift_start
shift_type
BUS_SCHEDULE
normal_eta_mins
arrival_time
departure_time
day_type
schedule_status
schedule_id
SCHEDULED_ON
RUNS
HAS_STOP
BUS_TRIP_STOP
dynamic_eta
is_active
scheduled_arrival
stop_sequence
scheduled_departure
STOPS_AT
BUS_FARE_RULE
min_distance_km
max_distance_km
fare_rule_id
child_fare
normal_fare
student_fare
senior_fare
effective_from
CAUSES_DELAY
BUS_DELAY_EVENT
delay_minutes
reason
resolution_status
reported_at
delay_id
GENERATES
INVOLVED_IN
BUS_INCIDENT
incident_id
incident_type
severity_level
incident_datetime
status
TRACKED_BY
BUS_LIVE_TRACKING
tracking_id
latitude
longitude
speed_kmph
recorded_at
updated_eta
UNDERGOES
BUS_MAINTENANCE_LOG
log_id
maintenance_type
maintenance_date
status
actual_cost
BUS_ALERT
alert_id
alert_type
message
severity
is_resolved
metro.STATION
(External Reference)
CONNECTS METRO
CONNECTS BUS
INTERCHANGE_POINT
walking_dist_m
interchange_id
walking_time_min
connection_type
is_covered
is_active
PLANS
COMBINED_JOURNEY
from_location
to_location
journey_id
total_fare
journey_type
total_eta_mins
HAS_LEG
COMBINED_JOURNEY_LEG
from_stop_id
mode
leg_sequence
to_stop_id
leg_fare
schedule_ref_id
leg_eta_mins
BOOKS
BUS_TICKET
ticket_id
price_paid
status
valid_to
qr_code
booking_channel
PAID_VIA
BUS_PAYMENT
amount
paid_at
status
payment_id
payment_method
SCANNED_AT
BUS_STOP_SCAN
scan_role
scan_id
is_valid
device_id
scan_timestamp
ON_SCHEDULE
TRAVELLING_IN
BUS_TRAVELLING_IN
exited_at
boarded_at
status
HOLDS_PASS
BUS_TRAVEL_PASS
price
pass_type
valid_to
pass_id
qr_code
valid_from
FEEDBACK_FOR
GIVES
BUS_FEEDBACK
feedback_id
category
rating
comment
status
BUS_SYSTEM_CONFIG
config_key
config_value
description
data_type
metro.PASSENGER
(External Reference)
1
N
M

HOW TO JOIN WITH METRO ER DIAGRAM:
1. Copy this entire diagram (Ctrl+A → Ctrl+C)
2. Open metro ER diagram in draw.io
3. Paste (Ctrl+V) — it adds to same canvas
4. Move bus diagram to the right of metro diagram
5. DELETE the two red 'External Reference' boxes
(metro.STATION and metro.PASSENGER)
6. Draw lines to the ACTUAL metro entities:
• INTERCHANGE_POINT.metro_station_id → metro.STATION
• BUS_TICKET.passenger_id → metro.PASSENGER
• BUS_TRAVEL_PASS.passenger_id → metro.PASSENGER
• BUS_TRAVELLING_IN.passenger_id → metro.PASSENGER
• BUS_FEEDBACK.passenger_id → metro.PASSENGER
• COMBINED_JOURNEY.passenger_id → metro.PASSENGER
7. That's it — full integrated ER is ready!